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SQL Unit 1: Database Fundamentals

1. What is Data?

Data is a collection of raw facts and figures.

Examples:

- Vivek
- 21
- 95%

These values by themselves are data.

2. What is a Database?

A database is an organized collection of related data stored electronically.

Example:

Student Table

ID	Name	Marks
1	Vivek	90
2	Rahul	85

Interview Answer:

A database is an organized collection of data that allows efficient storage, retrieval, and management of information.

3. What is DBMS?

Full Form

Database Management System

A software used to create, manage, and manipulate databases.

Examples:

- MySQL
- Oracle

- PostgreSQL
- SQL Server

Interview Answer:

DBMS is software that helps users create, manage, retrieve, and update data in a database.

4. Functions of DBMS

- Data Storage
 - Data Retrieval
 - Data Security
 - Data Backup
 - Data Management
-

5. Advantages of DBMS

Reduced Data Redundancy

Duplicate data is minimized.

Better Security

Access control can be applied.

Data Consistency

Data remains accurate.

Data Sharing

Multiple users can access data.

Backup & Recovery

Data can be restored if lost.

6. What is RDBMS?

Full Form

Relational Database Management System

Stores data in the form of tables and establishes relationships between them.

Examples:

- MySQL
- PostgreSQL
- Oracle
- SQL Server

7. Difference Between DBMS and RDBMS

DBMS	RDBMS
Stores data	Stores data in tables
No relationships required	Supports relationships
Less secure	More secure
Small systems	Large systems

Interview Answer:

DBMS stores data, whereas RDBMS stores data in related tables using keys and relationships.

8. What is a Relational Database?

A database where data is stored in tables and tables are connected using keys.

Example:

Students Table

StudentID	Name
1	Vivek

Orders Table

OrderID	StudentID
101	1

StudentID creates relationship.

9. Examples of RDBMS

- MySQL
- PostgreSQL
- Oracle
- Microsoft SQL Server

SQL Unit 2: Database Structure & Keys

1. What is a Table?

A table stores data in rows and columns.

Example:

Students

ID	Name	Marks
1	Vivek	90

Interview Answer:

A table is a database object used to store data in rows and columns.

2. What is a Row?

A row represents one complete record.

Example:

ID	Name	Marks
1	Vivek	90

Entire row is one record.

3. What is a Column?

A column represents a specific attribute.

Example:

ID	Name	Marks
----	------	-------

ID, Name, Marks are columns.

4. What is a Field?

A field is a single value inside a row and column.

Example:

ID	Name	Marks
1	Vivek	90

"Vivek" is a field.

5. What is Schema?

Schema is the logical structure of a database.

It defines:

- Tables
- Columns
- Relationships
- Constraints

Interview Answer:

Schema is the blueprint of a database that defines how data is organized.

6. What is an Instance?

The actual data stored in a database at a particular moment.

Schema = Structure

Instance = Data

KEYS

7. What is a Key?

A key is an attribute used to identify records uniquely and establish relationships between tables.

8. What is Primary Key?

A column that uniquely identifies each record.

Properties:

- Unique
- Cannot be NULL

Example:

StudentID	Name
1	Vivek
2	Rahul

StudentID is Primary Key.

Interview Answer:

■ A Primary Key uniquely identifies each record in a table and cannot contain NULL values.

9. What is Foreign Key?

A column that creates a relationship between two tables.

Students Table

StudentID	Name
1	Vivek

Orders Table

OrderID	StudentID
101	1

StudentID in Orders table is Foreign Key.

Interview Answer:

A Foreign Key is a column that references the Primary Key of another table to establish a relationship.

10. What is Candidate Key?

A column that can become a Primary Key.

Example:

RollNo	Email
101	vivek@gmail.com

Both can uniquely identify records.

Both are Candidate Keys.

11. What is Composite Key?

A key made from multiple columns.

Example:

StudentID	CourseID
-----------	----------

Together they uniquely identify records.

12. What is Unique Key?

Ensures unique values.

Difference:

- Primary Key → Only one per table
- Unique Key → Multiple allowed

13. Difference Between Primary Key and Foreign Key

Primary Key	Foreign Key
Uniquely identifies records	Creates relationship
Cannot be NULL	Can contain NULL
One table's identifier	References another table

SQL Unit 3: SQL Fundamentals

1. What is SQL?

Full Form

Structured Query Language

SQL is a language used to communicate with databases.

Using SQL, we can:

- Create databases
- Create tables
- Insert data
- Update data
- Delete data
- Retrieve data

Interview Answer:

SQL is a standard language used to store, retrieve, manage, and manipulate data in relational databases.

2. Why is SQL Used?

- Store data
- Retrieve data
- Update data
- Delete data
- Manage databases

Example:

```
SELECT * FROM Students;
```

3. What are SQL Commands?

SQL commands are instructions used to perform operations on a database.

Example:

```
SELECT * FROM Students;
```

4. Types of SQL Commands

SQL commands are divided into:

DDL

Data Definition Language

DML

Data Manipulation Language

DQL

Data Query Language

DCL

Data Control Language

TCL

Transaction Control Language

5. What is DDL?

Full Form

Data Definition Language

Used to define database structure.

Commands:

- CREATE
- ALTER
- DROP
- TRUNCATE

Interview Answer:

DDL commands are used to create and modify database structures.

6. What is DML?

Full Form

Data Manipulation Language

Used to manipulate data.

Commands:

- INSERT
- UPDATE
- DELETE

Interview Answer:

DML commands are used to insert, update, and delete records from a database.

7. What is DQL?

Full Form

Data Query Language

Used to retrieve data.

Command:

```
SELECT
```

Interview Answer:

DQL is used to retrieve data from a database using SELECT statements.

8. What is DCL?

Full Form

Data Control Language

Used to control permissions.

Commands:

- GRANT
 - REVOKE
-

9. What is TCL?

Full Form

Transaction Control Language

Used to manage transactions.

Commands:

- COMMIT
 - ROLLBACK
 - SAVEPOINT
-

10. Difference Between DDL and DML

DDL	DML
Structure	Data
CREATE	INSERT
ALTER	UPDATE

DROP	DELETE
------	--------

SQL Unit 4: Database & Table Operations

1. What is CREATE DATABASE?

Used to create a new database.

Syntax:

```
CREATE DATABASE college;
```

Example:

```
CREATE DATABASE student_db;
```

2. What is CREATE TABLE?

Used to create a new table.

Syntax:

```
CREATE TABLE table_name(  
column1 datatype,  
column2 datatype  
);
```

Example:

```
CREATE TABLE Students(  
StudentID INT,  
Name VARCHAR(50),  
Marks INT  
);
```

3. What is ALTER TABLE?

Used to modify an existing table.

Examples:

- Add column
- Remove column
- Modify column

Example:

```
ALTER TABLE Students  
ADD Email VARCHAR(100);
```

4. Why Use ALTER TABLE?

When table structure needs changes after creation.

Example:

New field added later.

```
Phone_Number
```

5. What is DROP TABLE?

Deletes an entire table permanently.

Example:

```
DROP TABLE Students;
```

Result:

- Table deleted
 - Data deleted
-

6. What is TRUNCATE TABLE?

Deletes all records but keeps table structure.

Example:

```
TRUNCATE TABLE Students;
```

Result:

Before
100 rows

↓

After
0 rows

Table remains.

7. Difference Between DROP and TRUNCATE

DROP	TRUNCATE
Deletes table	Deletes data only
Structure removed	Structure remains

Cannot use table	Can use table again
------------------	---------------------

8. What is a Database Schema?

Schema is the structure of the database.

Contains:

- Tables
- Columns
- Keys
- Relationships

9. Example Table

Students Table

StudentID	Name	Marks
1	Vivek	90
2	Rahul	85

Here:

Table Name:

Students

Columns:

StudentID
Name
Marks

Rows:

1 Vivek 90

2 Rahul 85

10. Real Support Engineer Scenario

Question:

Customer says a new column is required in database.

Solution:

```
ALTER TABLE Students  
ADD Email VARCHAR(100);
```

11. Another Scenario

Question:

Need to remove all data but keep table.

Solution:

```
TRUNCATE TABLE Students;
```

SQL Unit 5: Data Manipulation (DML)

DML commands are used to **insert, update, and delete data** from tables.

1. What is DML?

Full Form

Data Manipulation Language

Used to manipulate records inside tables.

Commands:

- INSERT
- UPDATE
- DELETE

Interview Answer:

DML commands are used to insert, modify, and delete data stored in database tables.

2. What is INSERT?

Used to add new records into a table.

Syntax

```
INSERT INTO table_name  
VALUES(value1,value2,value3);
```

Example

```
INSERT INTO Students  
VALUES(1,'Vivek',90);
```

Result:

StudentID	Name	Marks
1	Vivek	90

3. Insert Specific Columns

```
INSERT INTO Students(StudentID,Name)  
VALUES(2,'Rahul');
```

4. What is UPDATE?

Used to modify existing records.

Syntax

```
UPDATE table_name  
SET column=value  
WHERE condition;
```

Example

```
UPDATE Students  
SET Marks=95  
WHERE StudentID=1;
```

5. Why is WHERE Important in UPDATE?

Without WHERE:


```
UPDATE Students  
SET Marks=95;
```

All rows get updated.

This is a common interview question.

6. What is DELETE?

Used to remove records from a table.

Syntax

```
DELETE FROM table_name  
WHERE condition;
```

Example

```
DELETE FROM Students  
WHERE StudentID=1;
```

7. DELETE Without WHERE

```
DELETE FROM Students;
```

Deletes all records.

Table structure remains.

8. Difference Between DELETE and TRUNCATE

DELETE	TRUNCATE
Removes rows	Removes all rows

Can use WHERE	Cannot use WHERE
Slower	Faster
Table remains	Table remains

9. Difference Between DELETE, DROP and TRUNCATE

DELETE	TRUNCATE	DROP
Deletes rows	Deletes all rows	Deletes table
WHERE possible	WHERE not possible	No WHERE
Structure remains	Structure remains	Structure deleted

Most Asked GreyOrange Question

DELETE vs DROP vs TRUNCATE

10. Real Support Engineer Scenario

Customer says:

Wrong marks entered for student.

Solution:

```
UPDATE Students
SET Marks=85
WHERE StudentID=1;
```

11. Another Scenario

Customer says:

Remove one record.

```
DELETE FROM Students  
WHERE StudentID=1;
```

SQL Unit 6: Data Retrieval & Filtering

This is the **most important SQL unit** for GreyOrange.

1. What is SELECT?

Used to retrieve data from a table.

Syntax

```
SELECT * FROM Students;
```

Example Output

StudentID	Name	Marks
1	Vivek	90
2	Rahul	85

Interview Answer:

SELECT is used to retrieve data from one or more tables.

2. Select Specific Columns

```
SELECT Name,Marks  
FROM Students;
```

3. What is WHERE?

Used to filter records.

Example

```
SELECT *  
FROM Students  
WHERE StudentID=1;
```

Returns only matching rows.

4. What is DISTINCT?

Removes duplicate values.

Example

```
SELECT DISTINCT City  
FROM Students;
```

5. What is ORDER BY?

Used to sort data.

Ascending

```
SELECT *  
FROM Students
```

```
ORDER BY Marks ASC;
```

Descending

```
SELECT *  
FROM Students  
ORDER BY Marks DESC;
```

6. What is LIMIT?

Used to restrict output.

Example

```
SELECT *  
FROM Students  
LIMIT 5;
```

Returns first 5 rows.

Filtering Operators

7. AND Operator

Both conditions must be true.

```
SELECT *  
FROM Students  
WHERE Marks > 80  
AND City='Delhi';
```

8. OR Operator

Any one condition can be true.

```
SELECT *  
FROM Students  
WHERE Marks > 80  
OR City='Delhi';
```

9. NOT Operator

Reverses condition.

```
SELECT *  
FROM Students  
WHERE NOT City='Delhi';
```

10. BETWEEN Operator

Checks values within a range.

```
SELECT *  
FROM Students  
WHERE Marks BETWEEN 70 AND 90;
```

11. IN Operator

Checks multiple values.

```
SELECT *  
FROM Students  
WHERE City IN('Delhi','Mumbai');
```

LIKE Operator (Very Important)

GreyOrange specifically mentioned this in the email.

12. What is LIKE?

Used for pattern matching.

Example

```
SELECT *  
FROM Students  
WHERE Name LIKE 'V%';
```

Finds names starting with V.

13. Wildcard %

Represents multiple characters.

Example

```
SELECT *  
FROM Students  
WHERE Name LIKE '%k';
```

Names ending with k.

14. Wildcard _

Represents one character.

Example

```
SELECT *  
FROM Students  
WHERE Name LIKE '_i%';
```

15. Search Text in Column

GreyOrange Example:

```
SELECT *  
FROM Users  
WHERE Name LIKE '%vivek%';
```

Interview Answer:

LIKE is used to search text patterns in a column using wildcards such as % and _.

16. Real Support Engineer Scenario

Customer says:

Search user named Vivek.

```
SELECT *  
FROM Users  
WHERE Name LIKE '%vivek%';
```

17. Another Scenario

Customer says:

Show students with marks above 80.

```
SELECT *  
FROM Students  
WHERE Marks > 80;
```


SQL Unit 7: Functions & Grouping

Functions perform calculations on data and GROUP BY is used to organize data into groups.

1. What are Aggregate Functions?

Aggregate functions perform calculations on multiple rows and return a single value.

Examples:

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

Interview Answer:

Aggregate functions perform calculations on a set of rows and return a single result.

2. COUNT()

Used to count records.

Example

```
SELECT COUNT(*)  
FROM Students;
```

Output:

```
50
```

Meaning:

50 students exist.

3. SUM()

Used to calculate total.

Example

```
SELECT SUM(Marks)
FROM Students;
```

Adds all marks.

4. AVG()

Used to calculate average.

Example

```
SELECT AVG(Marks)
FROM Students;
```

5. MAX()

Returns highest value.

Example

```
SELECT MAX(Marks)
FROM Students;
```

6. MIN()

Returns lowest value.

Example

```
SELECT MIN(Marks)
FROM Students;
```

7. What is GROUP BY?

Used to group rows having the same value.

Example Table

City	Students
Delhi	10
Delhi	15
Mumbai	20

Query:

```
SELECT City, COUNT(*)
FROM Students
GROUP BY City;
```

Output:

City	Count
Delhi	2
Mumbai	1

8. Why Use GROUP BY?

Used for:

- Reports
 - Analytics
 - Summaries
-

9. What is HAVING?

HAVING filters grouped data.

Example

```
SELECT City, COUNT(*)  
FROM Students  
GROUP BY City  
HAVING COUNT(*) > 5;
```

10. Difference Between WHERE and HAVING

WHERE	HAVING
Filters rows	Filters groups
Before GROUP BY	After GROUP BY
Cannot use aggregate functions	Can use aggregate functions

Interview Answer:

WHERE filters individual rows, whereas HAVING filters grouped results.

11. Real Support Engineer Scenario

Question:

Find how many orders belong to each customer.

```
SELECT CustomerID,  
COUNT(*)  
FROM Orders  
GROUP BY CustomerID;
```

SQL Unit 8: SQL Joins (Highest Priority)

GreyOrange specifically mentioned:

Difference between INNER JOIN and OUTER JOIN

This is one of the most important interview topics.

1. What is a JOIN?

A JOIN combines data from multiple tables based on a common column.

Interview Answer:

JOIN is used to retrieve data from two or more tables using a related column.

Example Tables

Students

StudentID	Name
1	Vivek

2	Rahul
3	Aman

Orders

OrderID	StudentID
101	1
102	2

2. What is INNER JOIN?

Returns only matching records from both tables.

Query

```
SELECT *
FROM Students
INNER JOIN Orders
ON Students.StudentID = Orders.StudentID;
```

Output

StudentID	Name	OrderID
1	Vivek	101
2	Rahul	102

Aman is not shown because he has no order.

3. What is LEFT JOIN?

Returns:

- All records from left table
- Matching records from right table

Query

```
SELECT *  
FROM Students  
LEFT JOIN Orders  
ON Students.StudentID = Orders.StudentID;
```

Output

StudentID	Name	OrderID
1	Vivek	101
2	Rahul	102
3	Aman	NULL

4. What is RIGHT JOIN?

Returns:

- All records from right table
- Matching records from left table

Query

```
SELECT *  
FROM Students  
RIGHT JOIN Orders  
ON Students.StudentID = Orders.StudentID;
```

5. What is FULL OUTER JOIN?

Returns:

- All records from both tables
- Matching wherever possible

Query

```
SELECT *  
FROM Students  
FULL OUTER JOIN Orders  
ON Students.StudentID = Orders.StudentID;
```

6. Difference Between INNER JOIN and LEFT JOIN

INNER JOIN	LEFT JOIN
Only matching rows	All rows from left table
Unmatched rows removed	Unmatched rows shown as NULL

7. Difference Between INNER JOIN and OUTER JOIN

INNER JOIN	OUTER JOIN
Matching data only	Matching + non-matching data
Fewer records	More records

This is directly mentioned in GreyOrange's FAQ.

8. What is NULL?

NULL means:

- No value
- Missing value

- Unknown value

Example:

```
OrderID = NULL
```

9. Real Support Engineer Scenario

Question:

Customer says order exists but customer details are missing.

You may use JOIN:

```
SELECT *  
FROM Customers  
INNER JOIN Orders  
ON Customers.CustomerID = Orders.CustomerID;
```

to investigate the issue.

SQL Unit 9: Constraints & Advanced SQL

1. What are Constraints?

Constraints are rules applied to columns to maintain data accuracy and integrity.

Interview Answer:

Constraints are rules enforced on table columns to ensure valid and consistent data.

2. PRIMARY KEY Constraint

- Uniquely identifies each record.
- Cannot contain NULL values.
- Only one Primary Key per table.

Example:

```
CREATE TABLE Students(  
  StudentID INT PRIMARY KEY,  
  Name VARCHAR(50)  
);
```

3. FOREIGN KEY Constraint

Creates relationship between tables.

Example:

```
CREATE TABLE Orders(  
  OrderID INT PRIMARY KEY,  
  StudentID INT,  
  FOREIGN KEY(StudentID)  
  REFERENCES Students(StudentID)  
);
```

4. NOT NULL Constraint

Column cannot contain NULL values.

Example:

```
CREATE TABLE Students(  
  Name VARCHAR(50) NOT NULL  
);
```

5. UNIQUE Constraint

Ensures all values are different.

Example:

```
CREATE TABLE Students(  
  Email VARCHAR(100) UNIQUE  
);
```

6. DEFAULT Constraint

Assigns default value if none is provided.

Example:

```
CREATE TABLE Students(  
  City VARCHAR(50) DEFAULT 'Delhi'  
);
```

7. What is NULL?

NULL means:

- Missing value
- Unknown value
- No value assigned

Example:

Name	Phone
Vivek	NULL

8. IS NULL

Find records having NULL values.

Example:

```
SELECT *  
FROM Students  
WHERE Phone IS NULL;
```

9. IS NOT NULL

Find records having values.

Example:

```
SELECT *  
FROM Students  
WHERE Phone IS NOT NULL;
```

10. What is UNION?

Combines results of multiple SELECT queries and removes duplicates.

Example:

```
SELECT Name FROM Students  
UNION  
SELECT Name FROM Teachers;
```

11. What is UNION ALL?

Combines results including duplicates.

Example:

```
SELECT Name FROM Students
UNION ALL
SELECT Name FROM Teachers;
```

12. Difference Between UNION and UNION ALL

UNION	UNION ALL
Removes duplicates	Keeps duplicates
Slower	Faster

13. What is a View?

A virtual table created from a query.

Example:

```
CREATE VIEW TopStudents AS
SELECT Name,Marks
FROM Students
WHERE Marks > 80;
```

Interview Answer:

A view is a virtual table based on the result of a SQL query.

14. Advantages of Views

- Simplifies queries
- Improves security
- Reusability

15. What is a Subquery?

A query inside another query.

Example:

```
SELECT Name
FROM Students
WHERE Marks >
(
SELECT AVG(Marks)
FROM Students
);
```

16. Why Use Subqueries?

- Complex filtering
- Reporting
- Data analysis

SQL Unit 10: SQL for Software Support

This unit is the most relevant to the **GreyOrange Software Support Engineer** role.

1. Why is SQL Important for Software Support?

Support Engineers use SQL to:

- Investigate customer issues

- Verify records
 - Search data
 - Troubleshoot database problems
-

2. Data Search Queries

Find all users:

```
SELECT *  
FROM Users;
```

3. Search Specific User

```
SELECT *  
FROM Users  
WHERE UserID = 101;
```

4. Search Text Using LIKE

GreyOrange specifically mentioned this.

Example:

```
SELECT *  
FROM Users  
WHERE Name LIKE '%vivek%';
```

Interview Answer:

LIKE is used to search text patterns in a database column.

5. Missing Data Investigation

Question:

Customer says order is missing.

Query:

```
SELECT *  
FROM Orders  
WHERE OrderID = 101;
```

6. Duplicate Data Investigation

Find duplicate emails:

```
SELECT Email,  
COUNT(*)  
FROM Users  
GROUP BY Email  
HAVING COUNT(*) > 1;
```

7. Customer Issue Investigation

Example:

Customer:

My account is not visible.

Check:

```
SELECT *  
FROM Users  
WHERE Email='user@gmail.com';
```

8. Database Connectivity Issue

Possible causes:

- Wrong credentials

- Database server down
 - Network issue
 - Application configuration issue
-

9. Root Cause Analysis (RCA)

Finding the actual reason behind a problem.

Example:

Issue:

■ User cannot login.

Investigation:

- Check user record
- Check logs
- Check database connection

Then identify the root cause.

10. Export Table to CSV

GreyOrange FAQ specifically mentioned this.

Example:

```
COPY(  
SELECT *  
FROM Students  
)  
TO '/path/file.csv'  
WITH CSV;
```

Used to export table data.

11. Common SQL Support Tasks

- Verify user records
 - Verify orders
 - Search logs stored in database
 - Check missing records
 - Check duplicate records
 - Generate reports
-

12. Real GreyOrange Interview Scenario

Scenario

Customer says:

My order is not visible.

Approach:

1. Search customer record

```
SELECT *  
FROM Customers  
WHERE CustomerID=101;
```

2. Search order record

```
SELECT *  
FROM Orders  
WHERE CustomerID=101;
```

3. Use JOIN if needed

```
SELECT *  
FROM Customers  
INNER JOIN Orders  
ON Customers.CustomerID=Orders.CustomerID;
```

4. Identify issue

5. Resolve or escalate